



ITU Electrical-Electronics Faculty – Electrical Engineering Department

ELECTRICAL DISTRIBUTION AUTOMATION LABORATORY

LAB REPORT

Experiment No and Title	Experiment 5 - Differential Relay
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Experiment date	21 / 05 / 2025
Submission date of the lab report	28 / 05 / 2025
Grade	

EXPERIMENT 5. DIFFERENTIAL PROTECTION RELAY

Protocol Sheet

This protocol paper will be filled in completely and delivered to the research assistant at the end of the experiment.

Date Of The Experiment	Lab. Session Day And Hour	Exp. Group
21.05.2025	Wednesday / 12:30	66

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$I_{diff} \geq 0.43$			$I_{diff} \gg 0.8$	
Trials	Fault Type	Operating Current	Values at the Instant of Tripping	
			I_{diff}/I_n	I_{rest}/I_n
1	↓	$0.95 I_n$	0.43	0.84
2	↑	$0.95 I_n$	0.43	1.71
3	↓	$1.5 I_n$	0.43	1.54
4	↑	$1.5 I_n$	0.66	2.64
5	↓	$1.85 I_n$	0.50	2.00
6	↑	$1.85 I_n$	0.80	3.30
7	↓	$2 I_n$	0.55	2.17
8	↑	$2 I_n$	0.80	3.45

1. Explain with your own words, the experiment procedure. Include the connection diagram, utilized equipment, rated quantities of the protected device, and all calculations you made and measurements you obtained.

Before starting the experiment, some information about the experiment was discussed, this part can be summarized as follows:

- The protection principle of a relay is based on the fact that it acts as a communication device during abnormal situations such as a short circuit fault.
- For over-current relay, time delay, preset value, minimum short circuit current and maximum rated current parameters are decisive.
- For a differential relay, a protected zone is defined and the direction as well as the amplitude of the current is critical. Therefore, a reference direction is determined.
- There is no time delay for the differential relay, it is instantaneous and more sensitive than the over-current relay and its selectivity feature is more prominent.
- The use of differential relay comes to the fore in faults caused by low amplitude current such as the transmission line touching the tree.

The basic structure of a differential relay is shown in Figure 1. As seen in the figure, two current transformers are used for the differential relay. The reason for using these transformers is to bring the current to values that the relay can read. Depending on the location of fault, there are various scenarios. To evaluate these situations, differential and restraining current values are defined. The formulas for calculating these currents are shown in (1) and (2) respectively.

$$I_{diff} = |I_1 + I_2| \quad (1)$$

$$I_{rest} = |I_1| + |I_2| \quad (2)$$

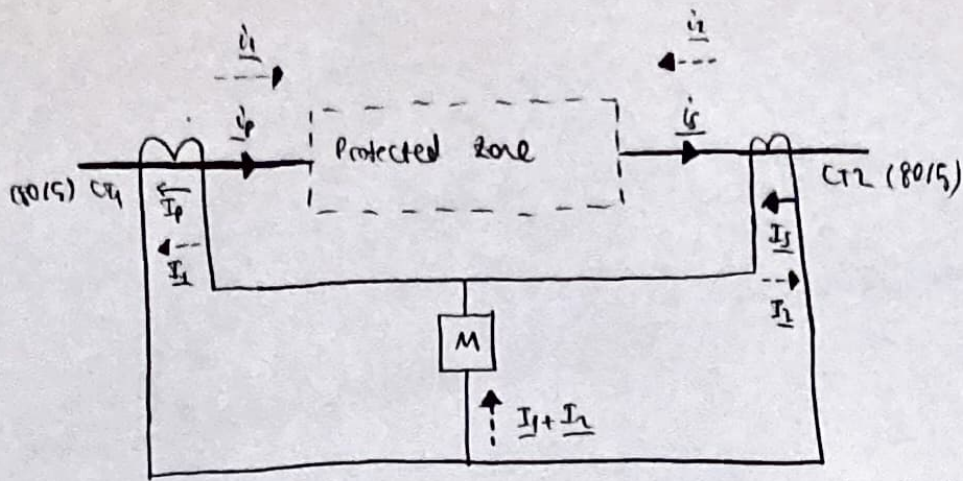
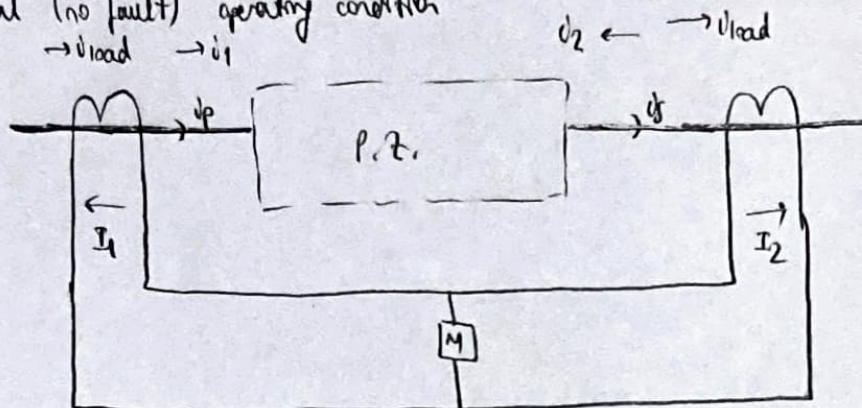


Figure 1. Basic connection scheme for differential relay

By comparing the differential and restraining current values, the relay evaluates the fault condition and sends a trip signal when the appropriate conditions are met. The four scenarios that may occur according to the fault condition are explained below:

① Normal (no fault) operating condition

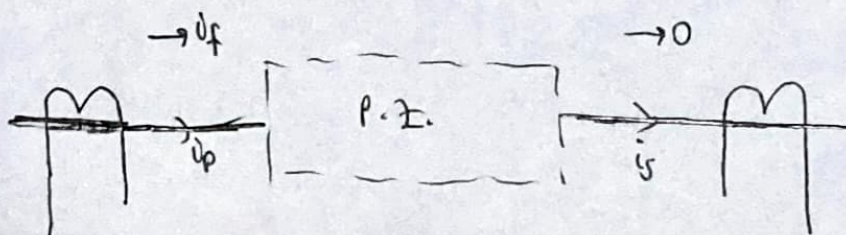


$$V_p = V_s = V_{load}, \quad I_1 = I_{load}, \quad I_2 = -I_{load}$$

$$I_{diff} = |I_{load} - I_{load}| = 0, \quad I_{rest} = |I_{load}| + |-I_{load}| = 2I_{load}$$

There is no fault in the system because it is a normal operating condition. Therefore, the relay does not send trip signal.

② Single-phase fault in the protected zone



$$V_p = V_f, \quad V_s = 0$$

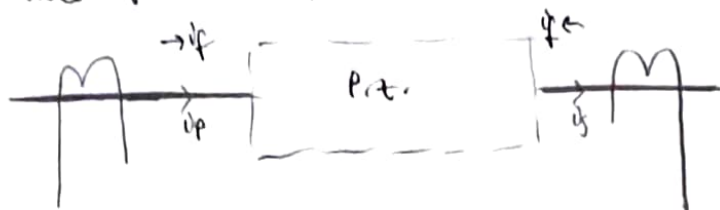
$$I_1 = I_f, \quad I_2 = 0$$

$$|I_{diff}| = |I_f + 0| = I_f$$

$$|I_{rest}| = |I_f| + |0| = I_f$$

There is a fault in the system because it is specified as "one-sided fault" when defining the condition at the beginning. Therefore, the relay sends a trip signal.

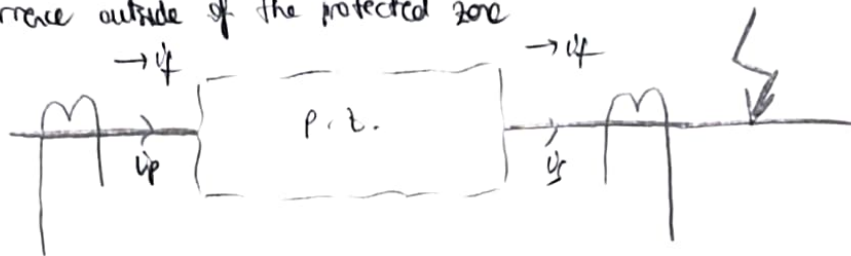
③ Double-sided fault in the protected zone



$$\begin{aligned} v_p &= v_f, \quad v_s = -v_f \\ I_1 &= I_f, \quad I_2 = I_f \\ I_{diff} &= |I_f + I_f| = 2I_f \\ I_{rest} &= |I_f| + |I_f| = 2I_f \end{aligned}$$

Similar to case 2, there is a fault in the system because the condition was initially defined as "double sided fault". Therefore, the relay sends a trip signal.

④ Fault occurrence outside of the protected zone



$$\begin{aligned} v_p &= v_f \\ v_s &= v_f \\ I_1 &= I_f \\ I_2 &= -I_f \end{aligned}$$

$$\begin{aligned} I_{diff} &= |I_f - I_f| = 0 \\ I_{rest} &= |I_f| + |-I_f| = 2I_f \end{aligned}$$

Although the condition of differential and restraining currents is the same as the first case, the relay does not send a trip signal because the fault occurs outside the protected zone. A fault occurring with high amplitude outside the protection zone causes saturation. This leads to incorrect measurements. Therefore, amplitude information is also important along with the direction information of current.

Each relay has a "tripping characteristic" associated with its operating principle. The tripping characteristic is determined according to current-time for overcurrent relay, R-X parameters for distance protection, differential and restraining currents for differential relay. However, per unit expressions of these values are used. This is done for simplification. It is more feasible to obtain a generalised graph instead of plotting individual graphs for each rated current condition.

In the experiment, one phase fault was investigated. For this, the values of

the required parameters were checked from the LL page. The LEDs used in the relay stand for pickup, trip and Buchholz trip respectively. The third LED is associated with pressure and humidity measurement. The primary and secondary side of the 3-phase transformer used in the experiment is 24.5 kV, 1 MVA, earthed and Y connected. The rated current is calculated as in (3). The basic connection diagram is given in Figure 2.

$$I = \frac{S}{\sqrt{3} V} = \frac{10^6}{\sqrt{3} (24.5 \times 10^3)} = 16.7 \text{ A} \quad (3)$$

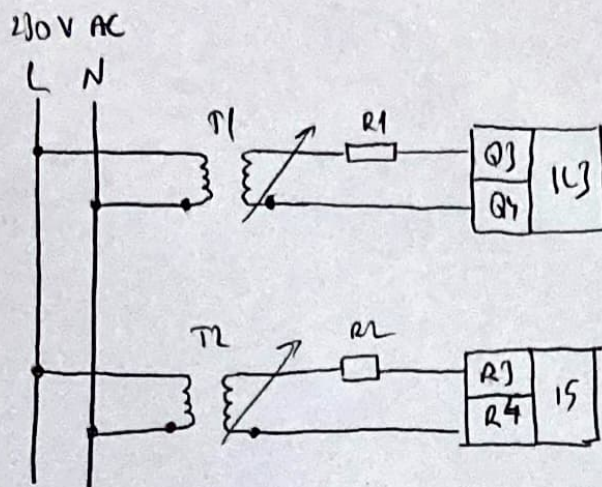


Figure 2. Connection diagram of the experiment

For differential current, the lower threshold is set to 0.4V and the upper threshold to 0.8. The conversion ratio for the current transformers used in the experiment is 10/5 as shown in Figure 1.

The situations examined in the experiment are described below. As observed in the experiment, when a fault occurs, the fault is first eliminated. The necessary statements about this situation were included during the stages of the experiment.

Trial 1: For the first fault type where the operating current value is 0.95 I_n , the reduction is examined. First, the currents for T1 and T2 were brought to approximately 95. Then the current on the T2 side was gradually decreased while the T1 side was kept constant and the first and second LEDs were turned on respectively. Then, the current was increased to eliminate the fault. After the LEDs were reset, Menu > Association > Trip log > Last fault tabs were followed and the value of "diff" and "rel" for LL were noted.

Trial 2: For the operating current in the first case, this time the error that occurs when the current is kept constant on one side and increased on the other side is analyzed. First, since it was performed immediately after the previous scenario, the "Menu" section was returned and the section with the % values of the currents was opened. Then, since the right side was not changed during the first case, only the left side was set to 95 and the current was increased. In the same way, the first and second LEDs lit up respectively. The current was decreased to eliminate the fault caused by the LEDs lighting up and after the trip signal was received, the new settings in the first case were repeated and the values were noted in order to read "diff" and "res" values.

In total, 8 cases were analyzed. Since the only parts changed for each case are fault type and operating current, the remaining steps can be summarized as follows with these values: For Trial 1, since the operating current was $1.5 I_n$, the current value of both sides was set to 150 and the current value of the left side was reduced. To eliminate the fault, the current was increased. For Trial 4, since the operating current was the same as in Trial 1, the left side current was brought back to 150 and then the current was increased. The current was decreased to eliminate the fault. For Trial 5 and 6, since the operating current was $1.85 I_n$, the right side remained constant at 185 while the left side was decreased and increased respectively after it was set to 185. For the elimination of the faults, increasing and decreasing operations were applied respectively. For Trials 7 and 8, since the operating current was $2 I_n$, the right side remained constant at 200, while the left side was ramped down and ramped up respectively. The measurement results are given in Table 1.

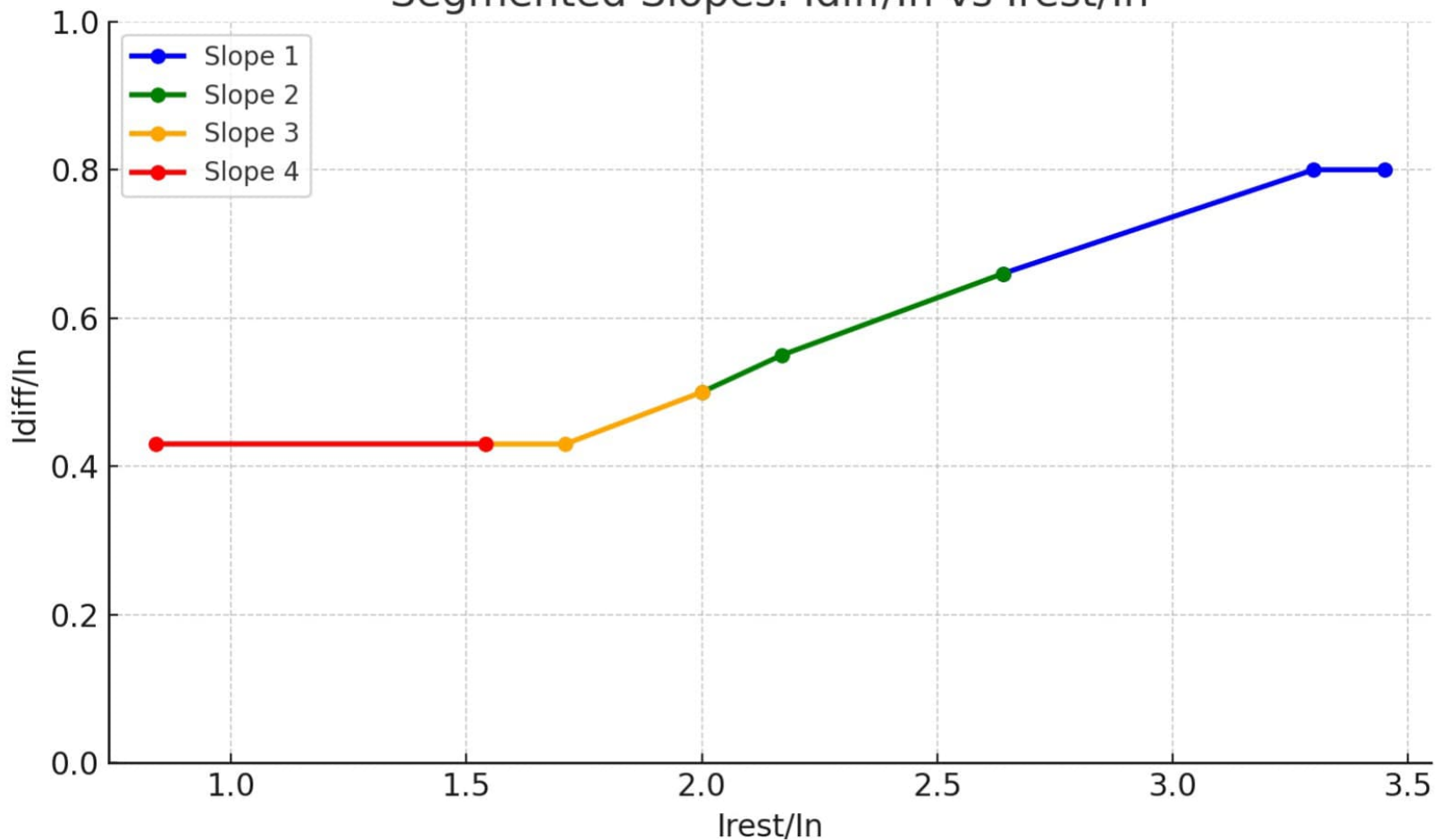
Table 1. Measurement results

Trials	$I_{diff} \geq 0.5$		$I_{diff} \geq 0.8$	
	Fault type	operating current	values at the instant of tripping I_{diff} / I_n	I_{res} / I_n
1	↓	$0.95 I_n$	0.43	0.84
2	↑	$0.95 I_n$	0.43	1.31
3	↓	$1.5 I_n$	0.43	1.85
4	↑	$1.5 I_n$	0.66	2.67
5	↓	$1.85 I_n$	0.50	2.00
6	↑	$1.85 I_n$	0.70	3.70
7	↓	$2 I_n$	0.55	2.17
8	↑	$2 I_n$	0.80	3.45

2. Plot and Evaluation of the Tripping Characteristic

$$\frac{I_{diss}}{I_n} = f\left(\frac{I_{rest}}{I_n}\right)$$

Segmented Slopes: Idiff/ln vs Irest/ln



Based on the experiment data, the tripping characteristic curve was plotted using Python by ordering the $\frac{I_{rest}}{I_n}$ values from largest to smallest. The curve represents the operational zones of the differential relay. As can be observed from the curve, when the normalized differential current $\frac{I_{diff}}{I_n}$ exceeds approximately 0.8, the point falls within the tripping (operating) zone. Conversely, for values below $\frac{I_{diff}}{I_n} < 0.45$, the relay remains in the restraining zone.

In the intermediate region $0.45 < \frac{I_{diff}}{I_n} < 0.9$, the relay's decision depends on the magnitude of the restraining current $\frac{I_{rest}}{I_n}$. Lower values of I_{rest} in this region tend to fall within the restraining zone, while higher values push the operation into the tripping zone. This slope-based segmentation is a key aspect of the relay's decision-making logic and ensures flexible response under varying load or fault conditions.

3. Harmonic Restraint and Inrush Current Characteristic

These features are implemented in transformer differential relays to improve reliability and avoid unnecessary tripping during non-fault transients.

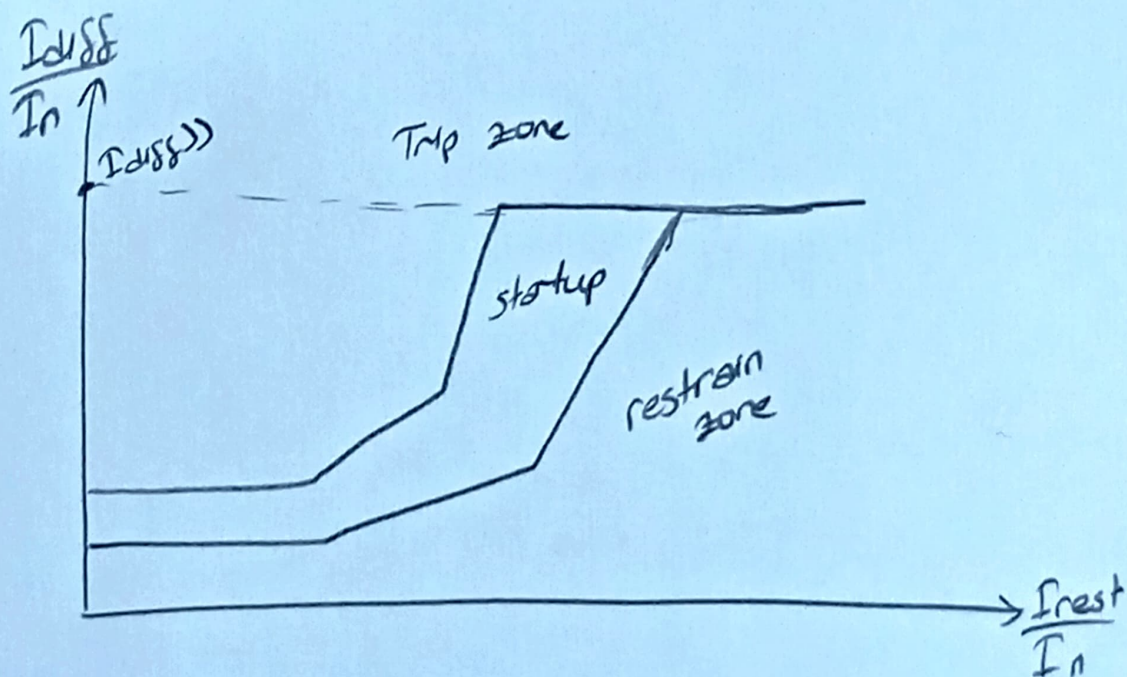
Harmonic Restraint

Primarily utilized in transformer protection. During initial energization, transformers can draw a high inrush current due to core magnetization. This inrush current often contains a high percentage of second harmonic components. The harmonic restraint function enables the relay to differentiate between genuine fault currents and harmless inrush conditions. By detecting elevated levels of second harmonic current, the relay inhibits tripping, thereby preventing false operation.

Startup Characteristic (Inrush blocking)

The startup characteristic, also known as the motor starting or inrush restraint zone, is particularly useful in systems that include motors. During motor startup, there is a transient where the differential current increases significantly due to initial torque demand, but this is not caused by an internal fault.

If not accounted for, this transient could trigger an unwanted trip. The startup zone, often shown graphically as a region with a more tolerant slope in the tripping characteristic, ensures that the relay restrains from issuing a trip signal during motor start, unless the current exceeds critical thresholds.



4. The relay you have used in the experiment is a transformer differential relay. What are the differences between a transformer differential relay and a line differential relay? Explain with your own words by integrating the references you have used.

Line differential protection offers significant benefits compared to distance protection, including superior performance in detecting remote faults, reliable operation during cross-country faults, and effective protection for short transmission lines. Additionally, it remains stable under conditions such as power swings, voltage reversals in series-compensated lines, mutual coupling with nearby lines, and load encroachment. Despite these strengths, line differential relays can still be vulnerable to issues like intense current transformer saturation, DDT network switching faults, with very high resistance, elevated capacitive currents, and communication failures [4.1].

Line differential protection operates based on Kirchhoff's current law, which asserts that, in the absence of a fault, the algebraic sum of currents entering the protected line is zero. Owing to this principle, the method provides high selectivity and isolates faults that occur specifically within the protected zone. When a fault is present within the protected section, the resulting operating current becomes sufficiently large to identify the fault and initiate a trip command for the affected line [4.2].

Line differential protection relies on an operational parameter known as the differential current (I_{diff}), which represents the absolute value of the total current entering the protected zone. To enhance the stability and reliability of the protection algorithm, a restraint current (I_{res}) is also computed by taking absolute value of each current entering the protected zone and adding all absolute values. The relay issues a trip signal only when I_{diff} surpasses both a predefined minimum threshold and a specific multiple of the restraint current [4.2].

A line differential protection system needs an individual relay installed at each terminal of the line being protected. For instance, in order to safeguard the transmission line between bus A and bus B, two separate devices are required. The device labeled "Diff A" monitors the current I_A and sends a trip signal to circuit breaker CB_B if an internal fault is detected. Similarly, the "Diff B" device measures the current I_B and triggers the circuit breaker CB_A in response to an internal fault [4.2].

[4.1] : <https://www.pacw.org/improvements-in-line-differential-relays>

[4.2] : J. Blumstein, T. Kerper and R. Mathussek, "Interoperability of Line Differential Protection," 2021 IEEE Conference for Protective Relaying Engineers (CPRE), College Station, TX, USA, 2021, pp. 1-6

Line differential protection algorithms require access to the primary current values at both ends of the protected line. However, in real-world applications, directly using primary currents is not feasible. Instead, current transformers are employed at each line end to convert the primary current into a secondary current suitable for measurement. These transformers inherently affect the accuracy of the current measurements used in the protection algorithm. To minimise discrepancies, it is advisable to use current transformers of the same class on both sides of the line, since any measurement mismatch will appear as a false differential current. In digital relay-based systems, the secondary current is digitised and transmitted to the opposite terminal. This digital processing - specifically, the analog-to-digital conversion and phasor computation - also introduces additional variations into the current measurements [4.1].

The percentage differential protection principle, first used for transformer protection, greatly enhanced the reliability of detecting external faults, especially under current transformer (CT) saturation conditions. One challenge for differential relays is the misoperation caused by transformer inrush currents, which are a result of transient magnetic flux changes. Early solutions to this issue involved introducing time delays or temporarily reducing the relay's sensitivity to cover the inrush period. Other methods proposed incorporating voltage signals either to restrain the relay or to supervise its operation. In modern systems, harmonic restraint and blocking are widely used to prevent incorrect tripping due to inrush or over-excitation, ensuring a high level of protection reliability. CT saturation, while a major source of false differential currents, affects internal and external faults differently - it can lead to unintended tripping on external faults or delay detection on internal ones if harmonic restraint is active. Initial designs handled this with delays or temporary desensitisation, but newer strategies use voltage-based logic or a combination of restraint and blocking for more effective transformer protection. Additionally, many relays now include an instantaneous overcurrent function to quickly trip down severe internal faults, even if CTs are saturated [4.1].

Differential protection is one of the fastest, most effective, and widely applied techniques for safeguarding power transformers. However, one of its major challenges lies in discriminating between transient phenomena such as inrush currents, over-excitation, and external faults - and genuine internal faults or winding defects. To address this, harmonic restraint techniques are

[4.1]: A. Goutas, J. Zocholl, G. Bernougal, H.J. Attive, "Performance analysis of traditional and improved transformer differential protective relays," 10th Annual Minnesota Power Systems Conference, 2020.

commonly used in modern differential protection systems. A more recent and advanced method involves the use of wavelet transform to separate current signals into approximation and detail coefficients. These coefficients are then analysed through specific criteria to accurately detect and differentiate faults. A long-standing difficulty in transformer protection is reliably telling apart magnetising inrush currents from internal fault currents, as both can appear similar and potentially lead to incorrect relay operation. To overcome this, some approaches analyse the trajectory of the differential current and apply weighted factors based on its position within the relay's characteristic curve. This enables clearer differentiation between internal faults and transient disturbances [4.4].

This type of protection is designed to detect winding-related issues, such as turn-to-turn short circuits or inter-winding faults, including phase-to-phase and three-phase faults. Additionally, it can be used to detect ground faults, especially in transformer installations where no direct earthing or grounding connection exists. The core difficulty in transformer differential protection remains the separation of inrush current from internal short-circuit current. Conventional methods rely on the presence of second harmonic components in inrush currents, using them as indicators to restrain relay operation. However, transformer energisation or fault produce high differential currents that mimic fault conditions, potentially triggering unnecessary tripping. Therefore, effective restraint mechanisms are essential, and the second harmonic restraint method remains the most widely adopted practical solution for identifying inrush conditions [4.5].

As a result, line differential protection and transformer differential protection are both protection methods based on the differential current principle, but differ in their application areas and challenges. Line differential protection detects faults within the line accurately and quickly by comparing currents at both ends of transmission lines; in these systems, high-speed and reliable communication is critical between relays at both ends of the transmission line. In addition, line protection takes special precautions against factors such as external influences and capacitive currents. Transformer differential protection, on the other hand, focuses on detecting faults in transformer windings and uses special algorithms (e.g. harmonic blocking) to prevent transient conditions, such as inrush current and harmonic components, from generating false alarms, especially due to the magnetic properties of the transformer.

[4.4] M. A. H. S. Kanak and M. S. Anwar, "Design and Implementation of Differential Relay for Power Transformer Protection," 2025 3rd International Conference on Advancements in Electrical and Electronic Engineering (ICAEEE), Coimbatore, India, 2025, pp. 1-6.

Furthermore, transformer-specific challenges such as CT saturation and magnetizing inrush are more pronounced in this type of protection. In summary, while line differential protection requires inter-crew coordination and communication of transmission lines, transformer differential protection requires more complex signal processing techniques to distinguish between transformer specific magnetic and electrical transients.

For the third question, "SIEMENS SIPROTEC 7UT6X User Guide Hayward, C.D. (1991). Harmonic current restrained relays for transformer differential protection" was used as reference.